Architecture

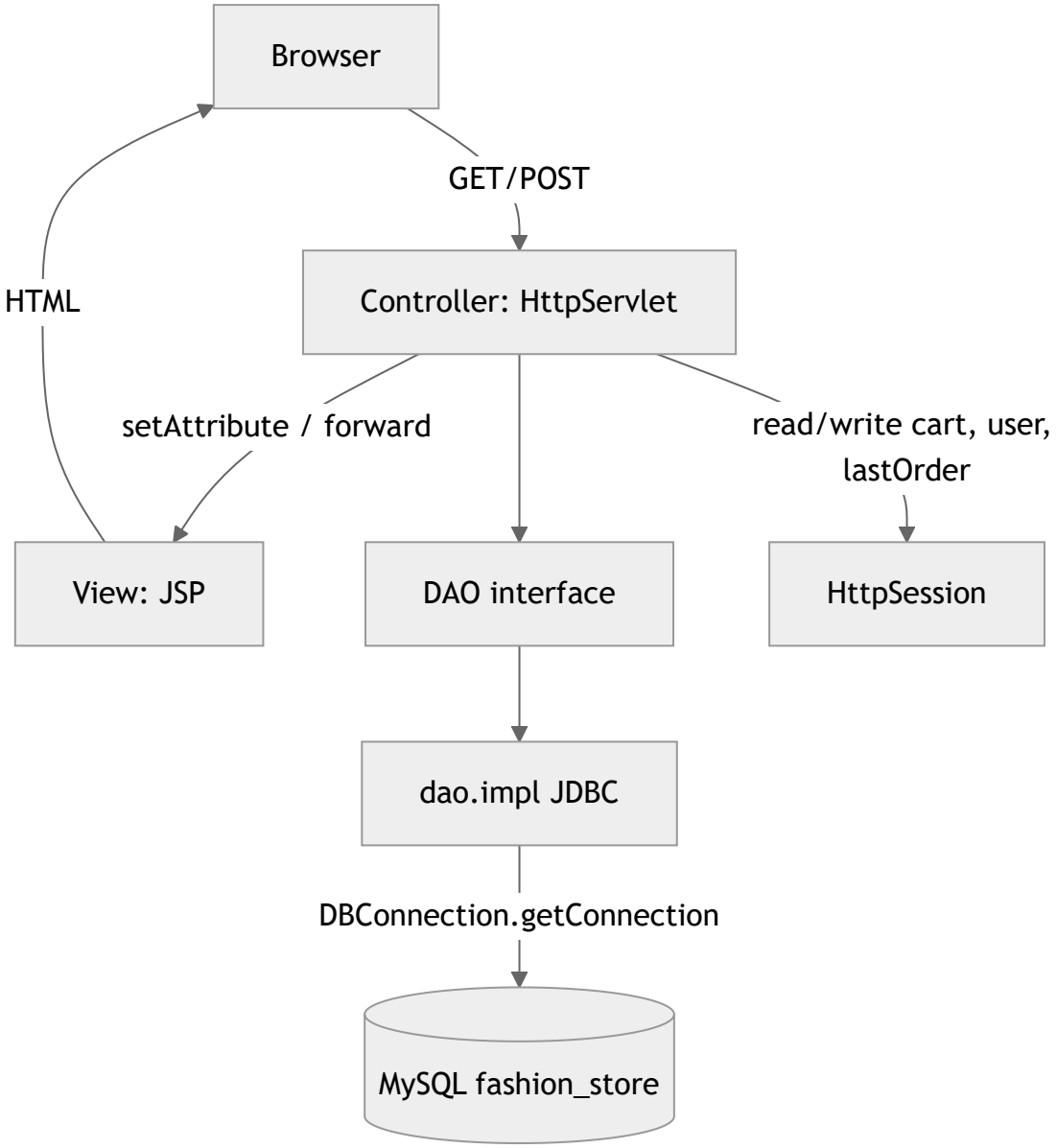
FashionStore is a **Jakarta Servlet MVC** WAR. HTTP hits annotated servlets (Controller). Servlets load POJOs via JDBC DAOs (Model) and forward to JSP (View). There is no Spring, no REST API, and no service/business layer between controllers and DAOs.

Package layout

```
com.fashionstore
├── controller   HttpServlet classes (@WebServlet)
├── model        Table-aligned POJOs
├── dao          DAO interfaces
├── dao.impl     JDBC implementations
└── util         DBConnection, AllDAOTest
```

Views live in `src/main/webapp/WEB-INF/views/` (not URL-addressable). Static CSS/images are under `src/main/webapp/assets/`. Welcome file `index.html` is listed in `web.xml`; app URLs are servlet mappings.

MVC

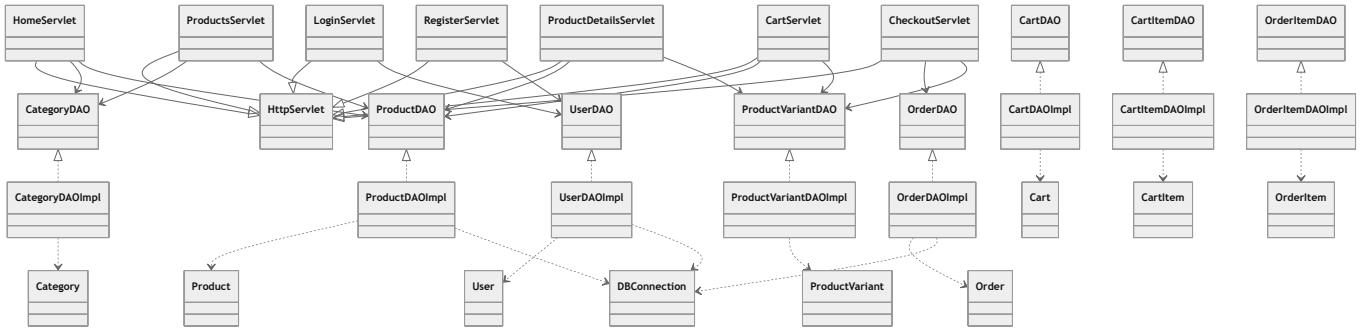


Request path:

1. Browser calls a mapped URL (`/home` , `/products` , `/cart` , …).
2. Servlet parses parameters and session.
3. Servlet calls one or more DAO methods (or only session, for cart display).
4. Servlet sets request attributes and **forwards** to a JSP, or **redirects** after a mutating POST.

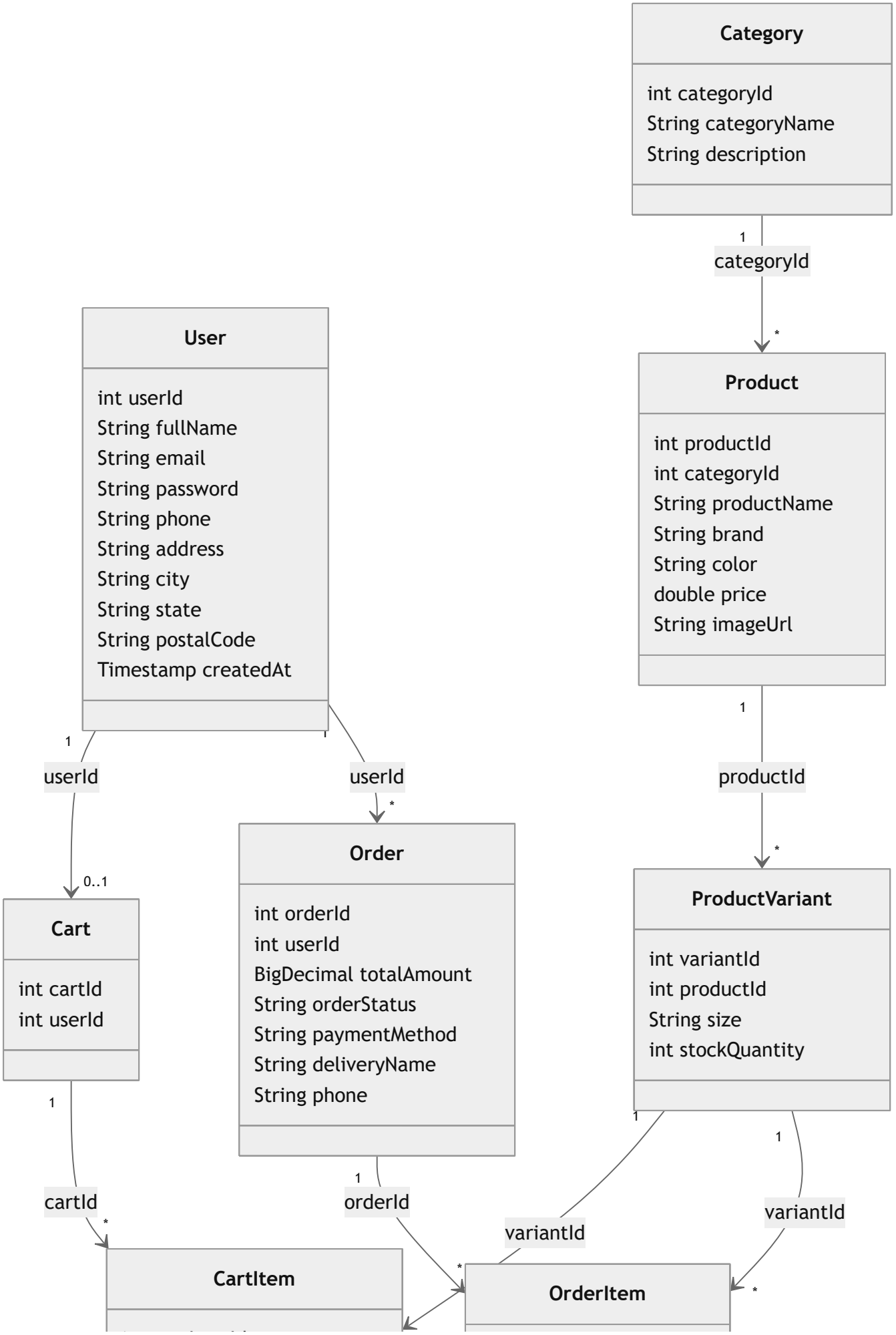
Session holds: `user` , `userId` , `cart` (`Map<String, CartItem>` keyed by `productId-variantId`), `lastOrder` , `lastOrderItems` .

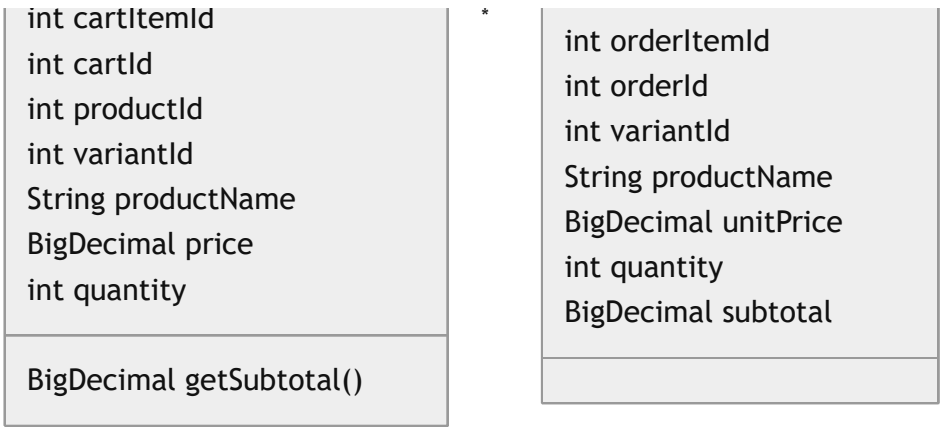
Layer class diagram



`CartDAO` / `CartItemDAO` / `OrderItemDAO` exist and are tested in `AllDAOTest`, but **no servlet currently injects them**. Live cart is session-only. Checkout writes the `orders` row only; line items stay in session as `lastOrderItems`.

Domain class diagram

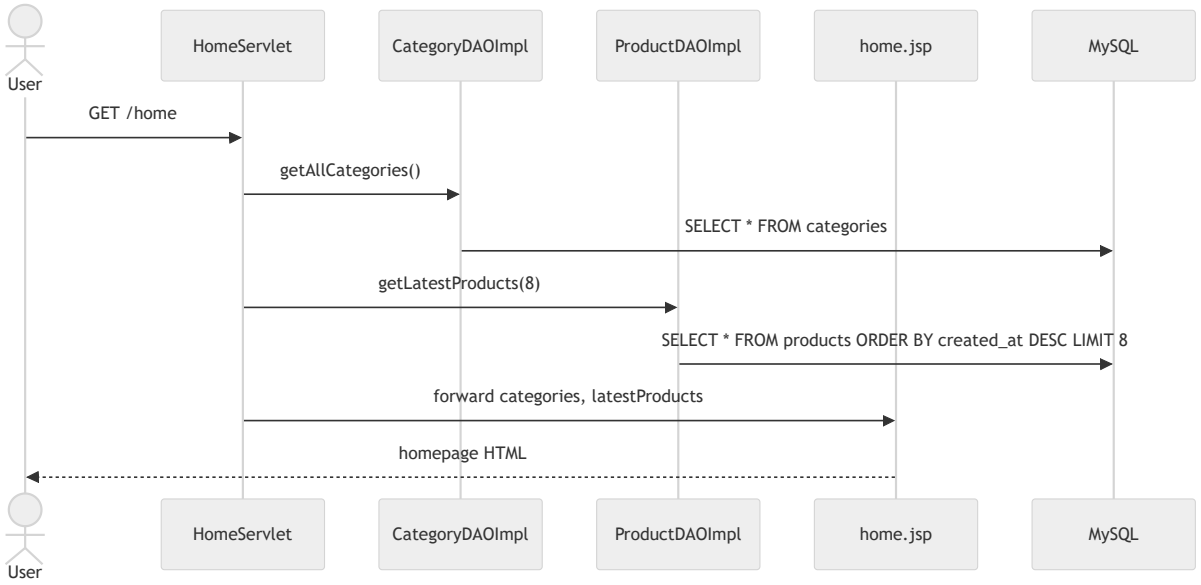

Category
int categoryId
String categoryName
String description
1
categoryId
*
User
int userId
String fullName
String email
String password
String phone
String address
String city
String state
String postalCode
Timestamp createdAt
Product
int productId
int categoryId
String productName
String brand
String color
double price
String imageUrl
1
productId
*
1
userId
1
userId
*
0..1
Cart
int cartId
int userId
Order
int orderId
int userId
BigDecimal totalAmount
String orderStatus
String paymentMethod
String deliveryName
String phone
ProductVariant
int variantId
int productId
String size
int stockQuantity
1
cartId
*
1
orderId
1
variantId
*
1
variantId
*
CartItem
OrderItem
*



`CartItem` used by the UI also carries display fields (`productName` , `brand` , `imageUrl` , `size` , `price`) that are **not** columns on `cart_items` . Those come from `Product` + `ProductVariant` when adding to the session cart.

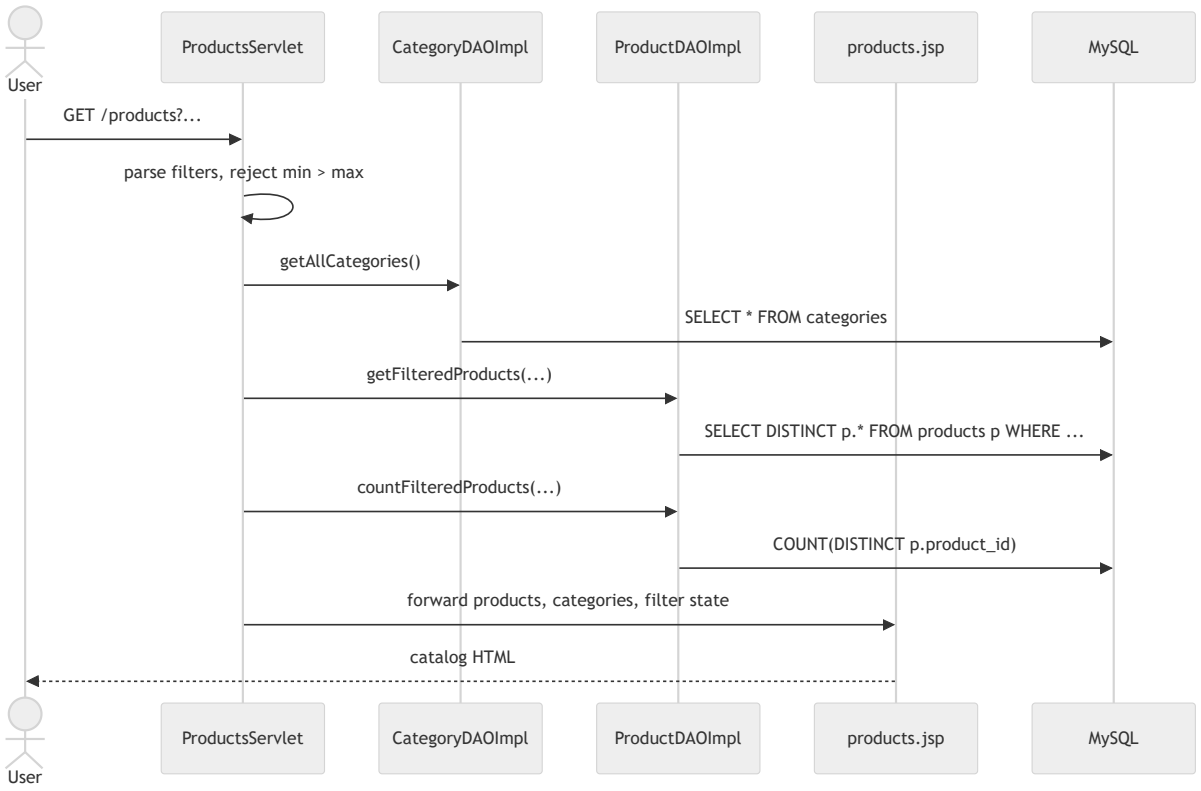
Sequence diagrams (live flows)

Home — `GET /home`



Catalog — `GET /products`

Query params: `keyword` , `categoryId` , `minPrice` , `maxPrice` , `sortBy` . Brand/color/size filters exist on the DAO but the servlet always passes `null` for those. Pagination is hard-coded to page 1, size 100.



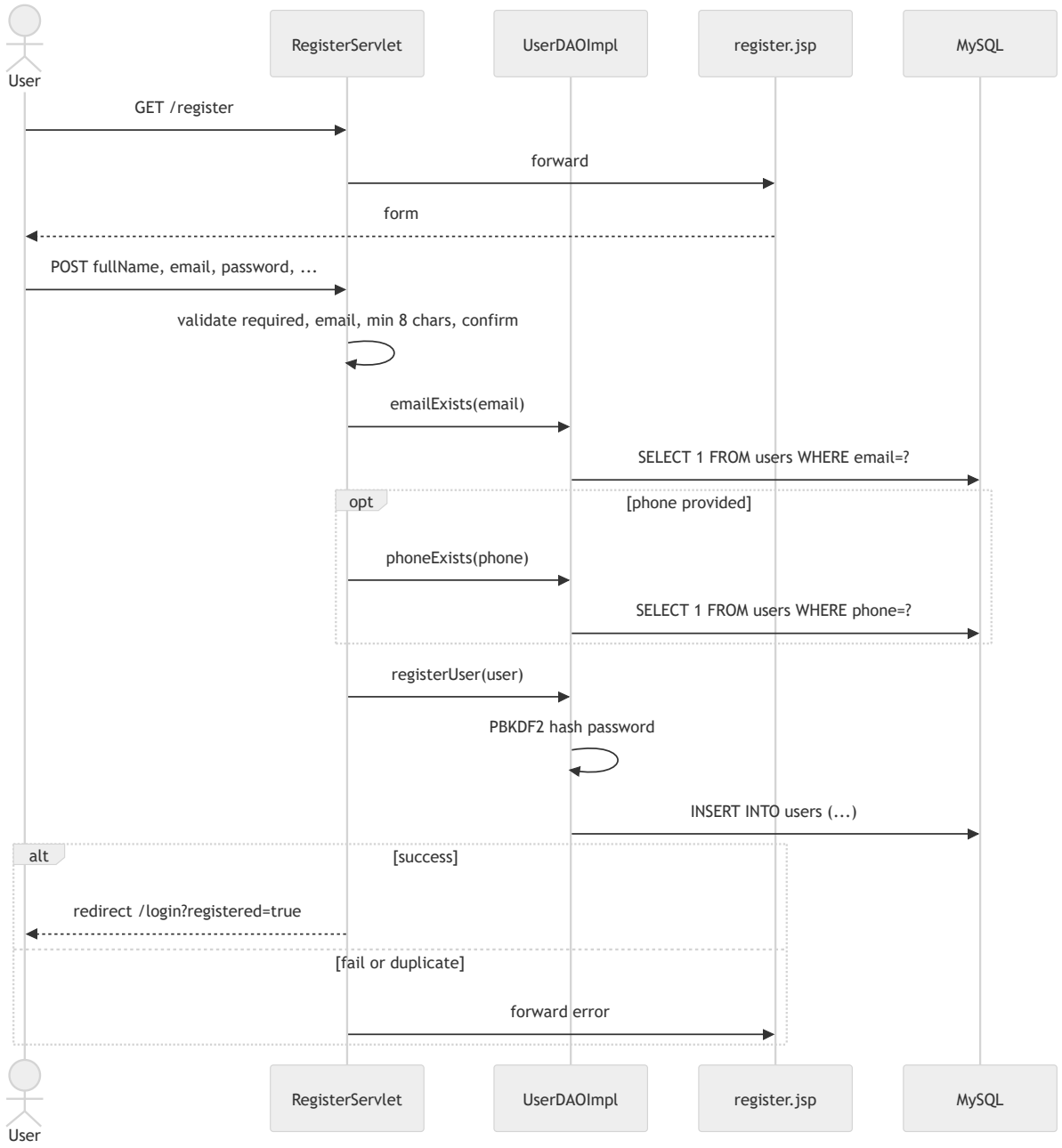
Navbar search is the same servlet: `GET /products?keyword=...`.

Product details — `GET /product?id=`

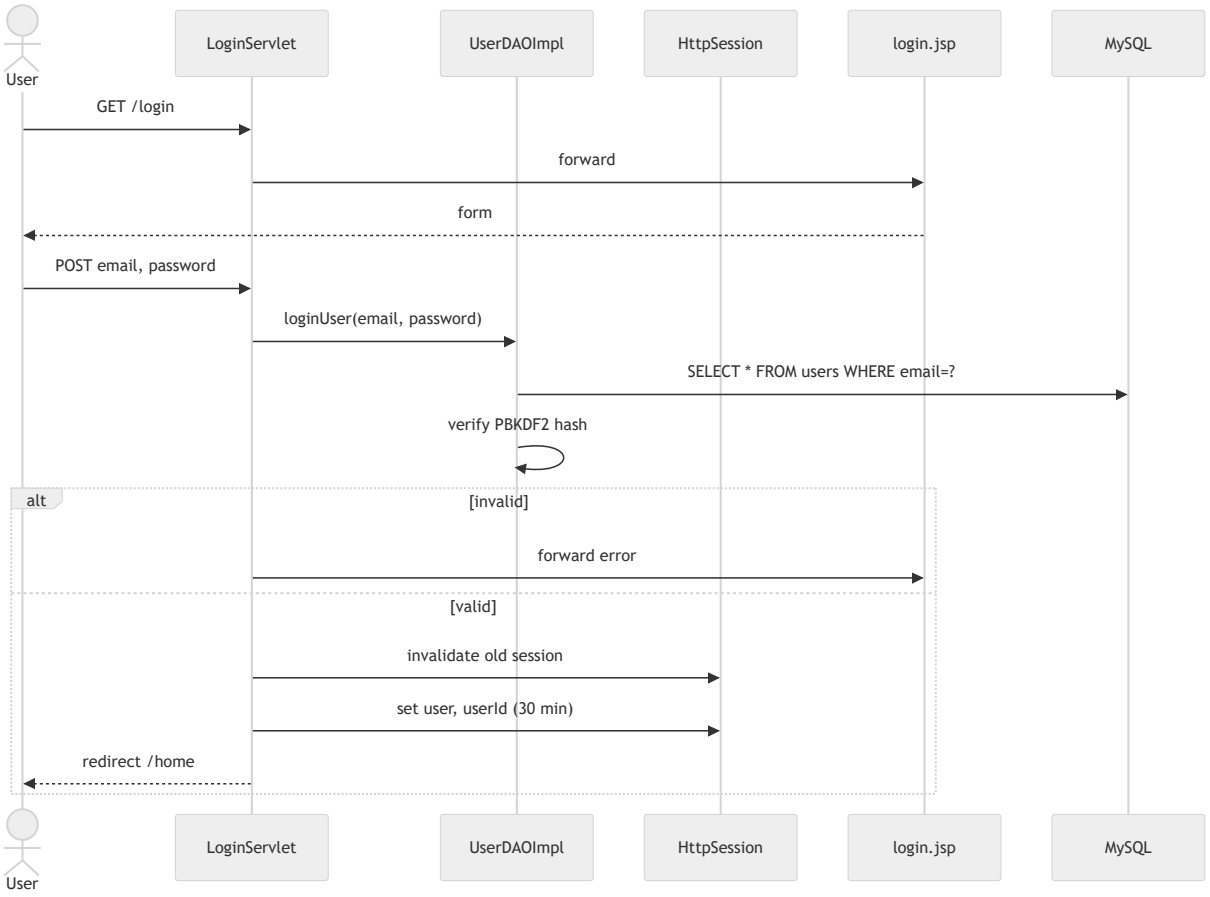


`getRelatedProducts` is on `ProductDAO` but is **not** called here.

Register — `GET/POST /register`



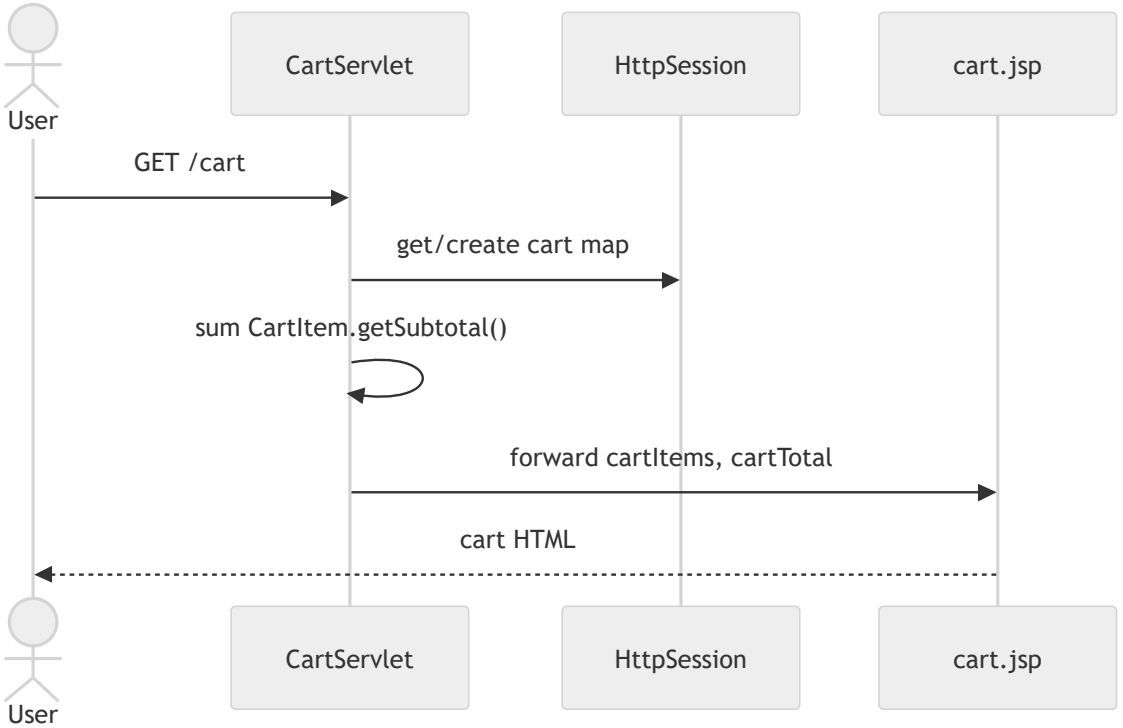
Login — `GET/POST /login`



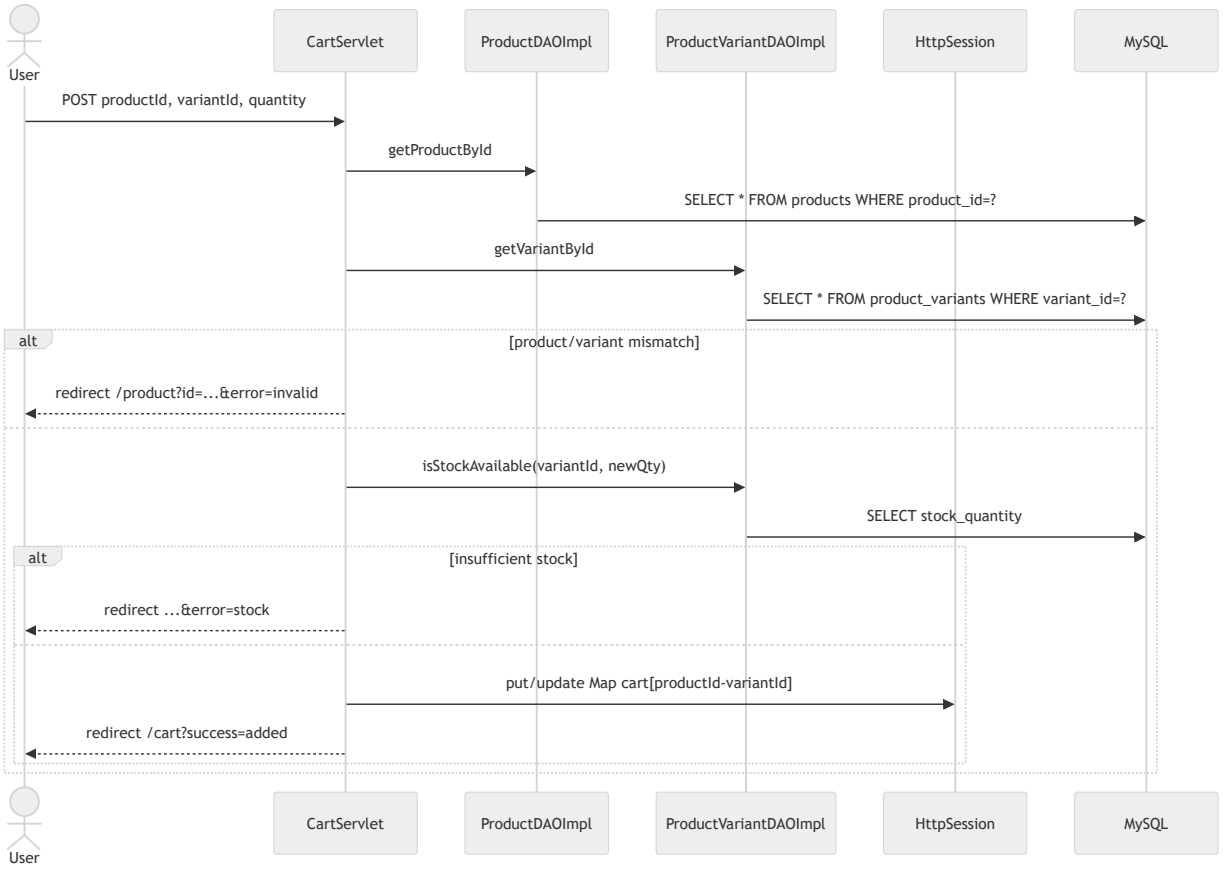
There is **no logout servlet**. Navbar always links to `/login`.

View cart — `GET /cart`

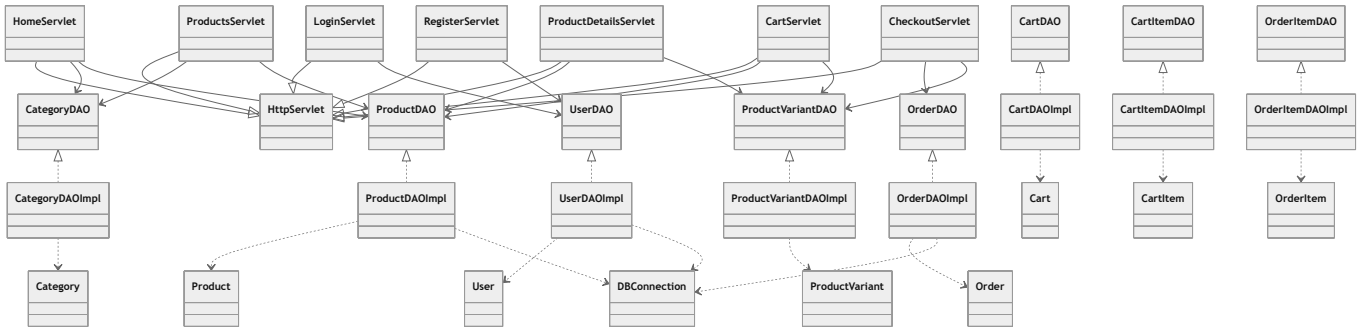
No DAO. Cart is a `LinkedHashMap` in session.



Add to cart — `POST /cart` `action=add`

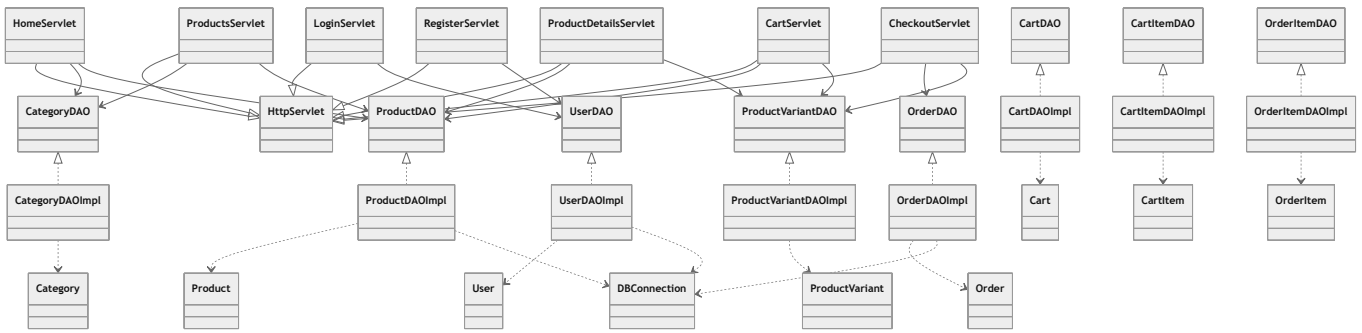


Update quantity — `POST /cart` `action=update`



0

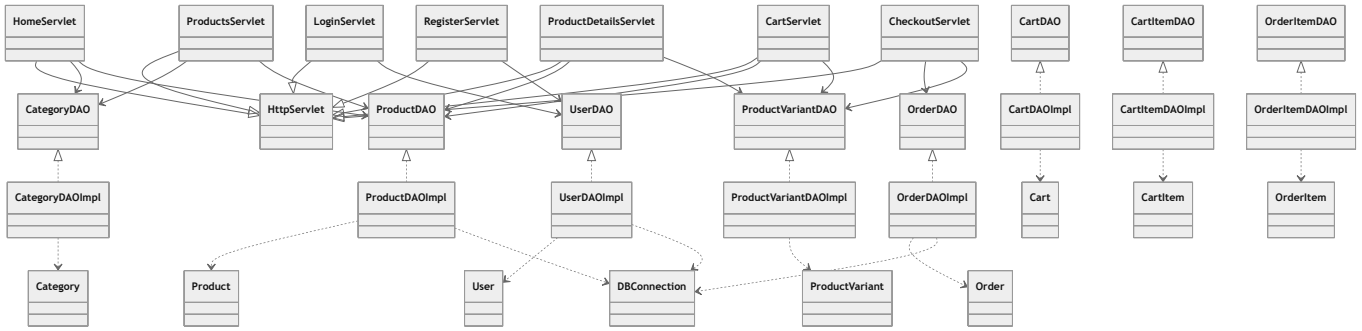
Remove line — `POST /cart` `action=remove`



1

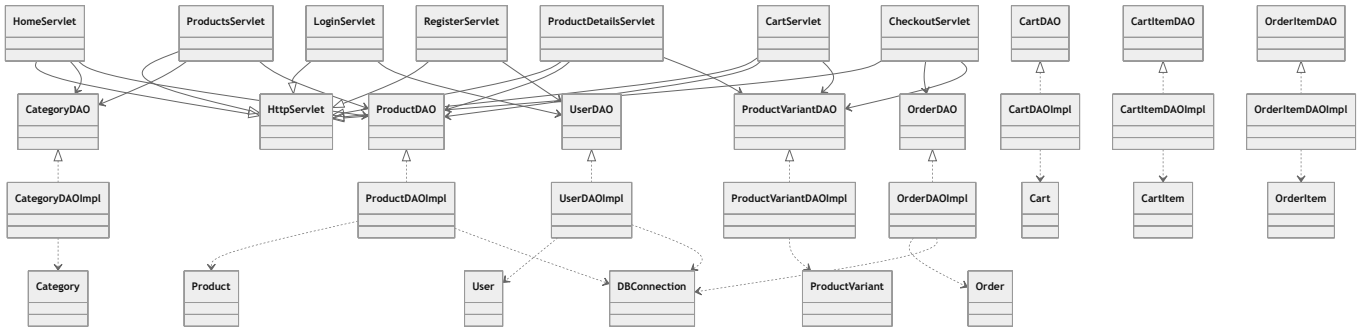
Stock is **checked**, never decremented, on add/update.

Checkout form — `GET /checkout`



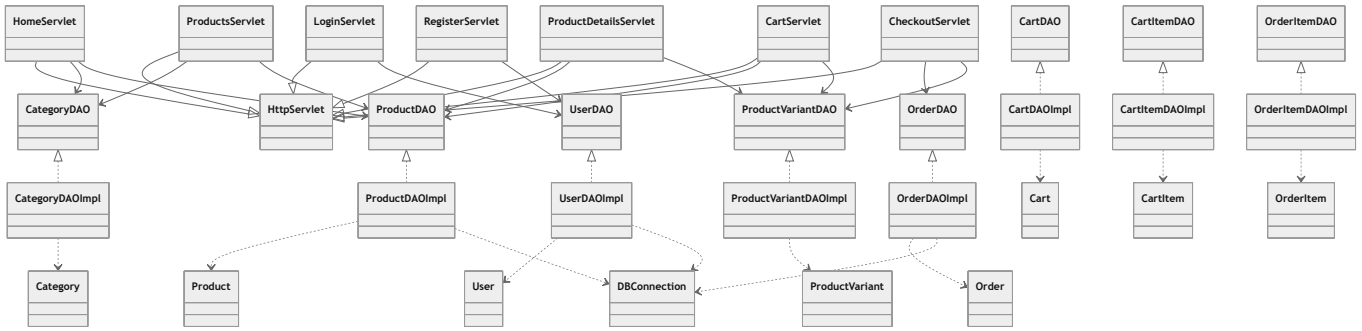
2

Place order — `POST /checkout`



3

Confirmation — `GET /checkout/confirmation`



4

URL → view

URL	Servlet	View
`GET /home`	`HomeServlet`	`home.jsp`
`GET /products`	`ProductsServlet`	`products.jsp`
`GET /product`	`ProductDetailsServlet`	`product-details.jsp`
`GET/POST /login`	`LoginServlet`	`login.jsp`
`GET/POST /register`	`RegisterServlet`	`register.jsp`
`GET/POST /cart`	`CartServlet`	`cart.jsp`
`GET/POST /checkout`	`CheckoutServlet`	`checkout.jsp`
`GET /checkout/confirmation`	`CheckoutServlet`	`order-confirmation.jsp`

Shared fragments: `partials/navbar.jsp`, `partials/footer.jsp`.